

3.3.6 Organic analysis

Our understanding of organic molecules, their structure and the way they react, has been enhanced by organic analysis. This section considers some of the analytical techniques used by chemists, including test-tube reactions and spectroscopic techniques.

3.3.6.1 Identification of functional groups by test-tube reactions

Content	Opportunities for skills development
The reactions of functional groups listed in the specification. Students should be able to: identify the functional groups using reactions in the specification.	AT b, d and k PS 2.2, 2.3 and 4.1 Students could carry out test-tube reactions in the specification to distinguish alcohols, aldehydes, alkenes and carboxylic acids.
Required practical 6 Tests for alcohol, aldehyde, alkene and carboxylic acid.	

3.3.6.2 Mass spectrometry

Content	Opportunities for skills development
Mass spectrometry can be used to determine the molecular formula of a compound. Students should be able to: use precise atomic masses and the precise molecular mass to determine the molecular formula of a compound.	

3.3.6.3 Infrared spectroscopy

Content	Opportunities for skills development
Bonds in a molecule absorb infrared radiation at characteristic wavenumbers. ‘Fingerprinting’ allows identification of a molecule by comparison of spectra. Students should be able to: use infrared spectra and the Chemistry Data Sheet or Booklet to identify particular bonds, and therefore functional groups, and also to identify impurities. The link between absorption of infrared radiation by bonds in CO₂, methane and water vapour and global warming.	Students should be able to use data in the Chemistry Data Sheet or Booklet to suggest possible structures for molecules.